

Forced-success mixture: concentration and scaled Laguerre tests

Independent companion calculation for the power lower-bound argument

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Let $p_1, \dots, p_m$ be a monotone uniform-interval comparison tensor with row weights w . Write $\delta_j = 1 - p_j$, $s = \sum_j \delta_j$, $\ell = \lfloor m/4 \rfloor$, $N = m - \ell$, and $\alpha = \ell/m$. Take m sufficiently large, so $1/5 \leq \alpha \leq 1/4$. Let B be a uniformly chosen ℓ -subset of the columns, $R = B^c$, and define

$$A_B = \prod_{j \in B} p_j, \quad \bar{A} = \mathbb{E}_B A_B, \quad x = \alpha s, \quad x_R = \frac{\ell}{N} \sum_{j \in R} \delta_j.$$

The finite mean-deficit balance theorem in `../rational_designs/asymptotic/finite_endpoint_odds_balance.tex` implies the global estimate

$$\max_j \delta_j \leq \frac{256(s+1)}{m}. \quad (1)$$

For $s \leq m/256$ this follows from that theorem's odds bound; for larger s it follows from $\delta_j \leq 1$.

Lemma 1 (Laplace mass). *For every fixed $c > 0$, $\mathbb{E}_w e^{-cs} \leq C_c/m$. Consequently $\mathbb{E}_w (1+s)^a e^{-cs} \leq C_{a,c}/m$ for every fixed nonnegative integer a .*

Proof. It suffices to treat $0 < c \leq 1$. On $s \leq m/1024$, (1) gives $\max_j \delta_j \leq 1/2$ for sufficiently large m . Choose any $k = \lfloor cm/1024 \rfloor$ columns. Their success product is at least

$$\exp \left(-2 \sum_{j \in I} \delta_j \right) \geq \exp(-512k(s+1)/m) \geq e^{-c(s+1)/2}.$$

Thus e^{-cs} on this region is at most $e^{c/2}$ times that success product. Its expectation is exactly $1/(k+1)$. On the complementary region $e^{-cs} \leq e^{-cm/1024}$. These two bounds prove the first assertion, after enlarging the constant to cover bounded m . The second follows by absorbing the polynomial into $e^{-cs/2}$. $\square$

Lemma 2 (Weighted mixture concentration). *At every row,*

$$\mathbb{E}_B x_R = x, \quad \mathbb{E}_B (x_R - x)^2 \leq C \frac{(s+1)^2}{m}, \quad \mathbb{E}_B [A_B |x_R - x|] \leq C e^{-\alpha s/2} \frac{s+1}{\sqrt{m}}. \quad (2)$$

If a differentiable function P satisfies

$$|P'(u)| \leq H(1+u)^a e^{\theta u} \quad (u \geq 0), \quad 0 < \theta \leq 1/16,$$

then

$$(\ell+1) \left| \mathbb{E}_w \mathbb{E}_B [A_B P(x_R)] - \mathbb{E}_w [\bar{A} P(x)] \right| \leq C_a H m^{-1/2}. \quad (3)$$

Proof. Sampling without replacement gives the exact variance

$$\text{Var}_B(x_R) = \frac{\ell^3}{N(m-1)} \left(\frac{1}{m} \sum_j \delta_j^2 - \frac{s^2}{m^2} \right).$$

Since $\sum_j \delta_j^2 \leq 256s(s+1)/m$, this proves the first two assertions in (2). Maclaurin's inequality gives $\bar{A} \leq (1-s/m)^\ell \leq e^{-\alpha s}$. As $A_B^2 \leq A_B$, Cauchy-Schwarz gives the last assertion in (2). Also $\max\{x, x_R\} \leq \alpha s/(1-\alpha) \leq s/3$. The mean value theorem, (2), and $\alpha \geq 1/5$ therefore bound the rowwise difference in (3) by

$$C_a H m^{-1/2} (1+s)^{a+1} e^{-\alpha s/2 + \theta s/3} \leq C_a H m^{-1/2} (1+s)^{a+1} e^{-s/16}.$$

The Laplace-mass lemma and $\ell+1 = O(m)$ prove (3). $\square$

Write $L_j(y) = \sum_{k=0}^j (-1)^k \binom{j}{k} y^k / k!$ for the Laguerre polynomials, and put

$$q_d(y) = \sum_{j=0}^{d-1} (1-j/d) L_j(y), \quad R_d(x) = \left(\frac{2}{\theta d} - x \right) q_d(\theta x)^2.$$

The standard Laguerre estimates $|L_j^{(k)}(y)| \leq \binom{j}{k} e^{y/2}$ for $y \geq 0$ imply

$$|R'_d(x)| \leq C_\theta d^3 (1+x) e^{\theta x}. \quad (4)$$

For $h = \lfloor d/8 \rfloor$ and $P_d(x) = h^{-2} (\sum_{j < h} L_j(\theta x))^2$ they similarly give $|P'_d(x)| \leq C_\theta d e^{\theta x}$. Thus (3) yields mixture-comparison errors $O(d^3/\sqrt{m})$ and $O(d/\sqrt{m})$ respectively.

Lemma 3 (Scaled taper under the exact beta law). *Let β_ℓ have density*

$$b_\ell(x) = \frac{\ell+1}{\ell}(1-x/\ell)^\ell \mathbf{1}_{[0,\ell]}(x).$$

For fixed $0 < \theta \leq 1/16$, sufficiently large d and ℓ ,

$$\int R_d d\beta_\ell \geq c_\theta > 0, \quad \int P_d d\beta_\ell \leq C_\theta/d. \quad (5)$$

Moreover $P_d(x) \geq 1/4$ whenever $0 \leq x \leq 2/(\theta d)$.

Proof. The ratio $b_\ell(x)/e^{-\theta x}$, extended by zero beyond ℓ , is decreasing, since its logarithmic derivative on the interior is $-1/(1-x/\ell)+\theta < 0$. Reweighting the probability measure proportional to $q_d(\theta x)^2 e^{-\theta x} dx$ by this decreasing ratio decreases its mean. The exact Laguerre identities give

$$\frac{\int_0^\infty x q_d(\theta x)^2 e^{-\theta x} dx}{\int_0^\infty q_d(\theta x)^2 e^{-\theta x} dx} = \frac{3}{\theta(2d+1)}.$$

It follows that

$$\int R_d d\beta_\ell \geq \frac{1}{2\theta d} \int q_d(\theta x)^2 d\beta_\ell.$$

On $0 \leq x \leq 1/(8\theta d)$, each $L_j(\theta x)$ with $j < d$ is at least $1/2$: its coefficient formula bounds the deviation from one by $e^{j\theta x} - 1 \leq e^{1/8} - 1 < 1/2$. Therefore $q_d(\theta x) \geq d/4$. For large d, ℓ , the beta density is bounded below by an absolute constant on this interval. The last integral is at least $c_\theta d$, proving the first claim.

For the second claim, $b_\ell(x) \leq 2e^{-\theta x}$ and Laguerre orthogonality give $\int P_d d\beta_\ell \leq 2/(\theta h) = O_\theta(d^{-1})$. If $x \leq 2/(\theta d)$ and $j < h \leq d/8$, the same coefficient estimate gives $|L_j(\theta x) - 1| \leq e^{1/4} - 1 < 1/2$. Thus the sum defining P_d is at least $h/2$, as claimed. $\square$

The exact beta law arises from the identity

$$(\ell+1)\mathbb{E}_w[A_B \prod_{j \in I} \ell(1-p_j)] = \int x^{|I|} d\beta_\ell(x), \quad I \subseteq B^c.$$

Consequently expectations of symmetric multiaffine polarizations on R are exact beta integrals. The comparison between a polarization and its scalar value at x_R is a separate required input; it is not proved in this companion note. Thus this note alone does not claim a power lower bound for the number of rows.